

10. Oxygen forms several molecular compounds of formula X_2O . These include compounds with fluorine and chlorine.

(a) Some thermodynamic data for F_2O and Cl_2O are given below.

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / \text{kJ mol}^{-1}$	Standard molar entropy, $S^\theta / \text{JK}^{-1} \text{mol}^{-1}$
F_2O	24.5	247
Cl_2O	80.3	266
O_2		205
F_2		203
Cl_2	0	

A student examines the data and states that there is sufficient information to find the minimum temperature for decomposition of F_2O to its elements but not for Cl_2O .

Is the student correct? Justify your answer.

[3]

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(b) F_2O is a strong oxidising agent and can oxidise Xe to XeF_4 whilst releasing oxygen gas.

(i) Write an equation for this process.

[1]

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(ii) Show that the xenon is oxidised in the process.

[1]

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- (c) F_2O reacts slowly with water to produce HF, a weak acid.



A small amount of F_2O is added to 400 cm^3 of water at 15°C and it reacts completely. The reaction produces 82 cm^3 of oxygen gas at this temperature and 1 atm pressure.

- (i) Calculate the concentration of the HF solution formed. [2]

concentration = mol dm^{-3}

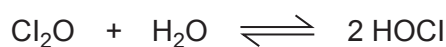
- (ii) Calculate the pH of the HF solution formed. [3]

$$K_a(\text{HF}) = 6.6 \times 10^{-4} \text{ mol dm}^{-3}$$

pH =



- (d) Cl_2O can react with water in a reversible reaction.



This reaction can occur in non-aqueous solvents and in a particular solvent the value of the equilibrium constant, K_c , at 273 K is 5.0 and it has no units.

- (i) Write an expression for the equilibrium constant K_c . [1]

- (ii) A student studies the equilibrium at a higher temperature. She adds samples of 0.40 mol of H_2O and 0.14 mol of Cl_2O to the non-aqueous solvent giving a total volume of 1000 cm^3 .

At equilibrium the concentration of HOCl is 0.22 mol dm^{-3} .

- I. Find the value of the equilibrium constant K_c at this temperature. [3]

$K_c =$

- II. State, giving a reason, whether this reaction is endothermic or exothermic. [2]

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III. HOCl can act as an oxidising agent.



Give the formulae of all the metal ions in the table that can be oxidised by HOCl (aq). Explain why you have chosen these ions. [2]

	Standard electrode potential, E^θ/V
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{Ti}^{3+} + 2\text{e}^- \rightleftharpoons \text{Ti}^+$	+1.25
$\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$	+1.69
$\text{Co}^{3+} + \text{e}^- \rightleftharpoons \text{Co}^{2+}$	+1.82

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